MATHEMATICS FOR EVERYONE

For Mathematics
and
For all my fellow Mathematicians

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01. Tips for Using Symbols in Mathematics

01. Never start a sentence with a symbol.

$x^2 - 6x + 8 = 0$ has two distinct roots. ✗

The equation $x^2 - 6x + 8 = 0$ has two distinct roots. ✓

02. Separate symbols with words, not punctuation, if possible.

With the exception of a, b is the only root of $(x - a)(x - b) = 0$. ✗

With the exception of a, the number b is the only root of $(x - a)(x - b) = 0$. ✓

03. Except when discussing logic, avoid writing the following symbols in your assignments.

$$\Rightarrow, \quad \forall, \quad \exists, \quad \ni, \quad \text{etc.}$$

04. Be careful about using *i.e.* and *e.g.*

i.e. means "that is" and *e.g.* means "for example". It is better not to use them in your assignments; writing the words out is preferable.

$\sqrt{-1}$ and $\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$ are not rational numbers, *that is*, i and e are not rational numbers. ✓

05. Write integers as words when they are used as adjectives, and use numerals for large, complicated numbers and exact mathematical values.

There are three roots. ✓

There are 1,527 roots. ✓

Let $n = 3$. ✓

Let $n =$ three. ✗

The interval is $[0, 3]$. ✓

The interval is $[0, \text{three}]$. ✗

06. Don't mix words and symbols improperly.

Every integer ≥ 2 is a prime or is composite. ✗

Every integer exceeding 1 is prime or composite. ✓

If $n \geq 2$ is an integer, then n is prime or composite. ✓

Since $(x-2)(x-3) = 0$, it follows that $x = 2$ or 3. ✗

Since $(x-2)(x-3) = 0$, it follows that $x = 2$ or $x = 3$. ✓

07. Avoid using a symbol in the statement of a theorem when it is not needed.

Theorem. Every bijective function f has an inverse. ✗

→ The theorem does not depend on what the function is called, so the symbol f provides no value.

Theorem. Every bijective function has an inverse. ✓

08. Explain the meaning of every symbol that you introduce.

09. Use common symbols properly.

m, n	integers
A, B	sets
P, Q, R	propositions
f, g, h	functions

10. Use consistent symbols, unless there is a reason to the contrary.

If x and y are even integers, then $x = 2a$ and $y = 2r$ for some integers a and b. ✗

If x and y are even integers, then $x = 2a$ and $y = 2b$ for some integers a and b. ✓

02. Tips for Using Expressions in Mathematics

01. Write short expressions within the text, and lengthy ones as a display.

If expressions are relatively short, then they should probably be written within the text of the proof or discussion. But if they are lengthy, then write the expression as a "display", on a line of its own.

For example, if we expand $(a+b)^6$, then we obtain $(a+b)^6 = a^6+6a^5b+15a^4b^2+20a^3b^3+15a^2b^4+6ab^5+b^6$. ✗

For example, if we expand $(a+b)^6$, then we obtain ✓

$$(a+b)^6 = a^6+6a^5b+15a^4b^2+20a^3b^3+15a^2b^4+6ab^5+b^6.$$

02. Display several expressions that are linked by equal signs or inequality symbols.

Since $n = 2k+1$, it follows that ✓

$$\begin{aligned} n^3+3n^2-n+4 &= (2k+1)^3+3(2k+1)^2-(2k+1)+4 \\ &= (8k^3+12k^2+6k+1)+(12k^2+12k+3)-(2k+1)+4 \\ &= 8k^3+24k^2+16k+7 \\ &= 2(4k^3+12k^2+8k+3)+1. \end{aligned}$$

03. When breaking a lengthy expression across lines, end each line with an operator.

If we have to write a lengthy expression in a text or paragraph and break it into two or more lines, then we should always end each line with an operator, and not start a line with an operator.

For example, if we expand $(a+b)^4$, then we obtain $(a+b)^4 = a^4+4a^3b+6a^2b^2+4ab^3+b^4$. ✓

04. Display and label an expression that you want to refer to later.

If we want to refer to an expression that has already appeared, then this expression should be written as a display and labelled.

Expanding $(a+b)^4$, we obtain ✓

$$(a+b)^4 = a^4+4a^3b+6a^2b^2+4ab^3+b^4 \tag{1}$$

Setting $a = b = 1$ in (1) gives $16 = 1+4+6+4+1$.

03. Tips for Using Words and Phrases in Mathematics

01. I, we, one, let's

I	reflects ego
we	standard mathematical practice
one	a bit awkward
let's	also good, but less professional

It is also possible to avoid these words altogether.

02. Clearly, obviously, of course, certainly

These and similar words turn the reader off if what is written is not evident to the reader. They can give the impression that the author is putting the reader down.

03. Any, each, every

This statement is true for any integer n. ✗

→ Does this mean that it is true for some integer n, or for every integer n? The word *any* is vague here.

This statement is true for every integer n. ✓

Instead of *any*, always use *each* or *every*.

04. Do not write "Since . . . , then . . . "

Use one of these forms instead: "If . . . , then . . . ", "Since . . . , it follows that . . . " or "Since . . . , we have . . . ".

If n^2 is even, then n is even. ✓

Since n^2 is even, it follows that n is even. ✓

Since n^2 is even, n is even. ✓

Since n^2 is even, then n is even. ✗

→ The same applies to *because*.

05. Therefore, thus, hence, consequently, so, it follows that, this implies that

These words are used to proceed from something we have just learned to something else that can be concluded from it. It is good to introduce some variety into your writing.

06. That, which

Both are correct to use in a statement, but they do different jobs. *That* introduces a restrictive clause; *which* does not introduce a restrictive clause.

07. Use good English.

08. Write complete sentences.

09. End each sentence with a period.

10. Capitalize the first word of each sentence.

11. Capitalize theorems and lemmas when they are numbered.

For example: Theorem 1.15, Lemma 4.11, Result 3.14.

12. Do not hyphenate words that begin with the prefix *non*.

For example: nonempty, nonnegative, nondecreasing, nonzero.

13. Watch out for commonly misspelled words.

commutative	corollary	parallel
complement	lemma	preceding
consistent	theorem	proceed
feasible	occurrence	

Chapter 04 | Logic

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01. Syntax and Semantics

Syntax and Semantics

Syntax is the alphabet, together with the rules to combine the symbols of the alphabet into well-formed formulas.
Semantics is the interpretation of well-formed formulas.

Example. The following example is by Noam Chomsky, from his study of linguistics.

1. *Furiously sleep ideas green colorless.*

   Syntactically, this sentence is not a well-formed sentence.

2. *Colorless green ideas sleep furiously.*

   This sentence does not make any sense, even though it is syntactically correct: something cannot be colorless and green at the same time.

So in mathematics we should avoid this type of mistake, and make our expressions both well formed syntactically and meaningful semantically. And that is where logical propositions come in.

02. Logical Propositions

Logical Proposition

A **logical proposition** is a well-formed statement that is either true or false, but not both.

Being *well formed* is a syntactical constraint, and being *either true or false, but not both* is a semantical constraint.

Example. Which of the following are logical propositions?

1. *Is* 6 *a composite number?*

   This statement is interrogative, but logical propositions are declarative. It is not a logical proposition.

2. *Factorise* 6 *into the product of its prime factors.*

   This is an imperative statement (it gives a command). It is not a logical proposition.

3. *If* 3*, then* 6 *is a composite number.*

   Here we have a syntactic error: "If 3" is not a complete sentence. It is not a logical proposition.

4. 6 *is a composite number.*

   This is a logical proposition. It is syntactically and semantically correct, and declarative.

5. 6 *is a prime number.*

   This is also a logical proposition. It is syntactically and semantically correct, even though its truth value is false.

Now we will be discussing two questions:

1. What are the valid symbols we can use? This is the *alphabet*.
2. How can we combine those valid symbols into well-formed expressions? These are the *rules of formation*.

Exercises

1. Consider the following expression:

"This sentence is false."

Is the sentence a mathematical proposition?

03. Syntax for Propositional Logic

The alphabet of propositional logic consists of the following.

1. **Propositional variables:** $P, Q, R, \ldots$

   Each propositional variable stands for a logical proposition (a well-formed statement that is either true or false, but not both).

2. **Logical connectives:** used to build more complex expressions by combining one or more propositional variables.

	Connective	**Symbol**	**Pronunciation**
unary	negation	$\neg$	"not"
binary	conjunction	$\wedge$	"and"
	disjunction	$\vee$	"or"
	conditional	$\rightarrow$	"if ..., then ..."
	biconditional	$\leftrightarrow$	"if and only if"

3. **Punctuation:** the parentheses ( ), used to clarify the structure of a more complex expression.

Exercises

1. Explain how each of the following sentences, when seen as formal sentences, can be built from atomic propositions and connectives.

(a) If I can see the sun, then I can't see the moon.

(b) I don't see the sun, neither do I see the moon.

(c) I go to bed only if I eat my dinner and brush my teeth.

(d) I play basketball unless it rains outside.

04. Well-Formed Formulas

The rules of formation tell us how to combine the alphabet into well-formed formulas.

Well-Formed Formula (WFF)

The well-formed formulas are defined recursively as follows.

1. Every propositional variable $P, Q, R, \ldots$ is a WFF.
2. If P is a WFF, then so is $(\neg P)$.
3. If P and Q are WFFs, then so are
$$(P \wedge Q), \quad (P \vee Q), \quad (P \to Q), \quad (P \leftrightarrow Q).$$
4. Nothing else is a WFF.

In $(P \to Q)$: when P is true, Q must be true.

Example.

WFFs	**Non-WFFs**
1. P	1. $\to P$
2. $P \to Q$	2. $P\neg Q$
3. $(\neg(P \leftrightarrow (\neg Q)))$	3. $)P \vee Q($

So far we have only discussed the syntactic aspect of propositional logic, that is, how to construct formulas correctly using the rules of WFFs.

But these rules do not guarantee that WFFs will always make sense semantically, that is, they do not guarantee that a WFF will only be true or false, and not both. For that we have to use the truth table.

05. Semantics for Propositional Logic

Our goal is to evaluate the truth value (T or F) of a WFF. Here is how we do it.

1. We give an interpretation to each atomic proposition. This means we assign a truth value to each propositional variable.
2. We extend the evaluation to the basic compound WFFs. This is the essential step: from the truth values of P and Q we get the truth values of
$$(\neg P), \quad (P \wedge Q), \quad (P \vee Q), \quad (P \to Q), \quad (P \leftrightarrow Q).$$

06. Truth Tables for the Basic Compound WFFs

P	$(\neg P)$
T	F
F	T

P	Q	$(P \wedge Q)$	$(P \vee Q)$	$(P \to Q)$	$(P \leftrightarrow Q)$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

- "If P, then Q" is false if and only if the antecedent P is true and the consequent Q is false.
- The biconditional is true when P and Q have the same truth value.

Disjunction is inclusive. This means $P \vee Q$ is true when P is true, or Q is true, or both P and Q are true.

Example. *You can have coffee or tea.*

Let P: "You can have coffee" and Q: "You can have tea". Then $P \vee Q$ allows:

	Mathematics	**English**
coffee only	✓	✓
tea only	✓	✓
coffee and tea	✓	×
neither	×	×

The conditional is vacuously true when P is false. $(P \to Q)$ says: if P happens, then Q must happen. If P never happened, then the condition was never triggered, and therefore it cannot contradict Q. Hence $(P \to Q)$ is vacuously true.

Example. Find the truth value of the proposition $(((P \vee Q) \wedge (\neg R)) \to S)$, given the interpretation that P is true, and Q, R, S are all false.

P	Q	R	S	$(P \vee Q)$	$(\neg R)$	$((P \vee Q) \wedge (\neg R))$	$(((P \vee Q) \wedge (\neg R)) \to S)$
T	F	F	F	T	T	T	F

So under this interpretation the proposition is false.

The WFF above involves 4 individual propositions, so there are $2^4 = 16$ different ways of assigning truth values to these individual atomic propositions.

Filling in a truth table. A strategy to fill in the values of each atomic proposition in each row: number all the proposition columns from right to left. Then

- in the 1st column, change the T/F value after every row;
- in the 2nd column, change it every 2 rows;
- in the 3rd column, change it every 4 rows;
- in the 4th column, change it every 8 rows.

Exercises

1. Find the complete truth table for the WFF

$$(((P \vee Q) \wedge (\neg R)) \to S).$$

2. Given propositions P, Q, R, S, W, is there a truth-value assignment which renders all the following propositions true simultaneously? Justify your answer.

(a) $((P \wedge R) \leftrightarrow S)$

(b) $(\neg(P \to (Q \vee W)))$

(c) $((S \to W) \wedge ((\neg R) \to (\neg W)))$

07. Tautology, Contradiction and Contingency

Tautology, Contradiction and Contingency

A proposition is a

- **tautology** if it is always true, under any interpretation;
- **contradiction** if it is always false, under any interpretation;
- **contingency** if it is true under some interpretation, and false under some other interpretation.

Tautologies and contradictions are usually compound propositions. But in some texts atomic propositions are named as

1. **verum** ($\top$), which is constantly true;
2. **falsum** ($\bot$), which is constantly false.

Example. Let P be an atomic proposition. Determine if each of the following propositions is a tautology, a contradiction, or a contingency.

1. $(P \vee (\neg P))$ 2. $(P \wedge (\neg P))$ 3. $(P \to (\neg P))$

P	$\neg P$	$(P \vee (\neg P))$	P	$\neg P$	$(P \wedge (\neg P))$	P	$\neg P$	$(P \to (\neg P))$
T	F	T	T	F	F	T	F	F
F	T	T	F	T	F	F	T	T
		tautology			contradiction			contingency

Exercises

1. Decide if the following logical sentences are tautologies, where P, Q, R are any logical propositions.

(a) $((P \to Q) \wedge ((\neg P) \leftrightarrow Q))$

(b) $(((P \to Q) \wedge (Q \to R)) \to (P \to R))$

2. So far, all the logical connectives that we have been working with are known as unary and binary connectives. This means that they take either one or two inputs and produce one output. We now consider the following ternary logic connective $\oplus(P, Q, R)$, which takes in three inputs and produces one output via the following table.

P	Q	R	$\oplus(P, Q, R)$
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	F
F	F	F	F

The operation returns true only if exactly two inputs are true. Rewrite the ternary operation using only $\wedge$, $\vee$, $\neg$, P, Q and R.

08. Logical Implication and Equivalence

Logical Implication and Logical Equivalence

Let P and Q be two propositions. We say that

- P **logically implies** Q, and denote it by $P \Rightarrow Q$, when $(P \to Q)$ is a tautology;
- P is **logically equivalent** to Q, and denote it by $P \Leftrightarrow Q$, when $(P \leftrightarrow Q)$ is a tautology.

Example.

- Does $(P \to Q)$ logically imply Q?
- Does $((P \to Q) \wedge P)$ logically imply Q?

P	Q	$P \to Q$	$(P \to Q) \wedge P$	$((P \to Q) \to Q)$	$(((P \to Q) \wedge P) \to Q)$
T	T	T	T	T	T
T	F	F	F	T	T
F	T	T	F	T	T
F	F	T	F	F	T

In the last row, $((P \to Q) \to Q)$ is false, so it is not a tautology, while $(((P \to Q) \wedge P) \to Q)$ is a tautology. Therefore

$$(P \to Q) \not\Rightarrow Q \qquad \text{and} \qquad ((P \to Q) \wedge P) \Rightarrow Q.$$

Example. Let P, Q be two propositions and consider the compound proposition $P \to Q$. Among the three propositions

$$1.\ (Q \to P), \qquad 2.\ (P \wedge (\neg Q)), \qquad 3.\ ((\neg Q) \to (\neg P)),$$

- which is logically equivalent to $(P \to Q)$?
- which is logically equivalent to $(\neg(P \to Q))$?

P	Q	$(P \to Q)$	$(\neg(P \to Q))$	$(Q \to P)$	$(P \wedge (\neg Q))$	$((\neg Q) \to (\neg P))$
T	T	T	F	T	F	T
T	F	F	T	T	T	F
F	T	T	F	F	F	T
F	F	T	F	T	F	T
				converse	negation	contrapositive

Comparing the columns,

$$(P \to Q) \Leftrightarrow ((\neg Q) \to (\neg P)) \qquad \text{and} \qquad (P \wedge (\neg Q)) \Leftrightarrow (\neg(P \to Q)).$$

So the contrapositive is logically equivalent to $(P \to Q)$, and $(P \wedge (\neg Q))$ is its negation. The converse $(Q \to P)$ is equivalent to neither.

Sufficient and necessary conditions. If $P \Rightarrow Q$, then we say that

1. P is a **sufficient** condition for Q, and
2. Q is a **necessary** condition for P.

Example. A new (binary) logical connective, denoted $\triangle$, has been discovered. Its full truth table is not known, but some properties of it have been observed.

P1: $(P \triangle P)$ is logically equivalent to $(\neg P)$.

P2: $(P \triangle Q)$ is not a contradiction.

P3: If P is true and Q is false, then $P \triangle Q$ is true.

Can the complete truth table be determined from these rules? If so, what is it? What are all the possible truth tables for it? Justify your answer.

By P1, $T \triangle T = \neg T = F$ and $F \triangle F = \neg F = T$. By P3, $T \triangle F = T$. P2 is then already satisfied, since $T \triangle F = T$. Nothing determines $F \triangle T$:

P	Q	$\neg P$	$P \triangle P$	$Q \triangle Q$	$P \triangle Q$
T	T	F	F	F	F
T	F	F	F	T	T
F	T	T	T	F	?
F	F	T	T	T	T

So the complete truth table cannot be determined. There are exactly two possible truth tables, one with $F \triangle T = T$ and one with $F \triangle T = F$.

Exercises

1. Verify that each of the following rules of inference is valid.

- **Modus ponens (MP):** from $(P \to Q)$ and P, derive Q.
- **Modus tollens (MT):** from $(P \to Q)$ and $(\neg Q)$, derive $(\neg P)$.
- **Hypothetical syllogism (HS):** from $(P \to Q)$ and $(Q \to R)$, derive $(P \to R)$.
- **Disjunctive syllogism (DS):** from $(P \vee Q)$ and $(\neg P)$, derive Q.
- **Addition (Add):** from P, derive $(P \vee Q)$.
- **Simplification (Simp):** from $(P \wedge Q)$, derive P.
- **Conjunction (Conj):** from P and Q, derive $(P \wedge Q)$.
- **Resolution (Res):** from $(P \vee Q)$ and $((\neg P) \vee R)$, derive $(Q \vee R)$.
- **Constructive dilemma (CD):** from $((P \to Q) \wedge (R \to S))$ and $(P \vee R)$, derive $(Q \vee S)$.
- **Destructive dilemma (DD):** from $((P \to Q) \wedge (R \to S))$ and $((\neg Q) \vee (\neg S))$, derive $((\neg P) \vee (\neg R))$.

2. You are attempting entry into MAT157 Island. Citizens of MAT157 Island can only say logical propositions. You are confronted by three citizens: Allen Iverson, Beyoncé, and Cauchy. Some citizens are mathematicians. The three citizens make the following statements:

Allen Iverson: "I am a mathematician."
Beyoncé: "I am not a mathematician."
Cauchy: "Allen Iverson and Beyoncé are not both mathematicians."

Assume furthermore that exactly one person is lying, and at least one of them is a mathematician. Determine how many mathematicians there are, and who is (or are) the mathematician(s).

3. Verify that each of the following rules of replacement is valid.

- **De Morgan's theorem (DM):** $(\neg(P \wedge Q)) = ((\neg P) \vee (\neg Q))$
- **De Morgan's theorem (DM):** $(\neg(P \vee Q)) = ((\neg P) \wedge (\neg Q))$
- **Commutation (Com):** $(P \vee Q) = (Q \vee P)$
- **Commutation (Com):** $(P \wedge Q) = (Q \wedge P)$
- **Association (Assoc):** $(P \vee (Q \vee R)) = ((P \vee Q) \vee R)$
- **Association (Assoc):** $(P \wedge (Q \wedge R)) = ((P \wedge Q) \wedge R)$
- **Distribution (Dist):** $(P \wedge (Q \vee R)) = ((P \wedge Q) \vee (P \wedge R))$
- **Distribution (Dist):** $(P \vee (Q \wedge R)) = ((P \vee Q) \wedge (P \vee R))$
- **Double negation (DN):** $P = (\neg(\neg P))$
- **Transposition (Trans):** $(P \to Q) = ((\neg Q) \to (\neg P))$
- **Material implication (Impl):** $(P \to Q) = ((\neg P) \vee Q)$
- **Material equivalence (Equiv):** $(P \leftrightarrow Q) = ((P \to Q) \wedge (Q \to P))$
- **Material equivalence (Equiv):** $(P \leftrightarrow Q) = ((P \wedge Q) \vee ((\neg P) \wedge (\neg Q)))$
- **Exportation (Exp):** $((P \wedge Q) \to R) = (P \to (Q \to R))$

09. Limitations of Propositional Logic

There are some limitations to propositional logic. The main one is the inaccessibility of the internal structure of a statement.

Example. Consider the statement *"Some cat chases Jerry."* In propositional logic the whole sentence is a minimal unit, a single proposition P. We do not know which cats these are, what "chases" means, or who Jerry is.

Example.

If Mochi is a cat, then Mochi is lovely.	$P \to Q$	
Mochi is a cat.	P	
Therefore, Mochi is lovely.	Q	✓

This inference is justified in propositional logic, by modus ponens.

Example.

Every cat is lovely.	H	
Mochi is a cat.	P	
Therefore, Mochi is lovely.	Q	×

- Although this inference sounds pretty reasonable, unfortunately we will not be able to justify its validity by using merely propositional logic.
- The reason is that the first statement is not of the form "if P, then Q". Thus we cannot apply modus ponens here.
- In the scope of propositional logic, we will not be able to tell what the meaning of "every" is, since the whole sentence is treated as a proposition in its own right.
- Also, just in the scope of propositional logic, we cannot explain how, if every cat is lovely, then Mochi is also lovely.

Example. *"Alice loves Bob"* and *"Bob loves Alice"*.

- From the frame of propositional logic, we do not know anything about the internal structure of these statements.
- What we mean is that propositional logic does not know anything about the statements (the way they are different, and so on). In *"Alice loves Bob"*, "Alice" is the subject, "loves" is the predicate and "Bob" is the object, but propositional logic does not know anything about this.
- The whole sentence is treated as an atomic proposition.

10. Predicate Logic

In propositional logic we do not have access to the internal structure of a given statement. To overcome this problem, we need to split the sentence into more detailed components, to recover the internal structure.

A sentence can be split into two components:

1. a **term** (the subject), and
2. a **predicate**.

For example, in *"Mochi is lovely"*, "Mochi" is the term and "is lovely" is the predicate.

Domain of Discourse, Terms and Predicates

- The **domain of discourse** D is the set of all objects of interest.
- A **term** is a symbol that is either a constant or a variable.
  - A **constant** stands for a specific element in D.
  - A **variable** can range over the elements in D.
- An **n-ary predicate** P is a symbol which becomes a formula when applied to n terms.

Example. Let the domain of discourse D be a set of cats, containing the cats c, m and a, and let P: "___ is lovely". Then

$$P(c) : c \text{ is lovely}, \qquad P(m) : m \text{ is lovely},$$
$$P(a) : a \text{ is lovely}, \qquad P(x) : x \text{ is lovely}.$$

Example.

1. P: "___ is an even number" is a unary predicate. For example, $P(4)$: 4 is an even number.
2. Q: "___ is divisible by ___" is a binary predicate. For example, $Q(x, y)$: x is divisible by y.
3. R: "max{___, ___} is greater than ___" is a ternary predicate.

In predicate logic, each sentence is considered as the result of applying an n-ary predicate to n terms. Also, the introduction of variables permits us to extend our study from a special case to the general:

- from a special case to a general case;
- from a single object to multiple objects;
- from a static point of view to a dynamical perspective.

Exercises

1. True or false: any predicate becomes a formula when applied to a single variable.
2. Identify the following sentences as formulas achieved by applying an n-ary predicate to term(s), where n is specified in [ ] after each sentence.
   - 8 is an even number. [1]
   - Toronto is in Canada. [2]
   - x is the greatest common divisor of a and 6. [3]
   - x is the integer part of π. [2]

11. Connectives and Predicate Logic

Connectives are used to generate compound formulas from atomic ones.

Example. Let

- $P(x)$: x is an even number;
- $Q(x, y)$: x is divisible by y;
- $R(x, y, z)$: $\max\{x, y\} > z$.

Then we can form, for example,

- $\neg R(x, y, z)$: $\max\{x, y\} \leq z$;
- $P(x) \to Q(x, y)$: if x is an even number, then x is divisible by y.

From the above we can say that connectives can be applied in predicate logic to form compound statements, just as we do with propositional logic.

But with some statements in predicate logic, like

Every cat is lovely,

the approach of using connectives fails.

- If D is small, like $D = \{\text{Chowi}, \text{Mochi}\}$, then the above statement can be constructed as

  $$P(\text{Chowi}) \wedge P(\text{Mochi}),$$

  that is, "Chowi is lovely and Mochi is lovely".
- But it becomes impossible to use connectives if D is the set of all the cats in the world.

And this is where quantifiers come to our rescue.

Exercises

1. Is each connective enough for the stated purpose?
   (a) Conjunction, to express that each of finitely many propositions holds true.
   (b) Conjunction, to express that each of infinitely many propositions holds true.
   (c) Disjunction, to express that one of finitely many propositions holds true.
   (d) Disjunction, to express that one of infinitely many propositions holds true.
2. Let $ABCD$ be a quadrilateral. Let the domain of discourse D consist of the four vertices A, B, C, D. Let P be a quaternary predicate, with

   $$P(x, y, t, s) : \|\overrightarrow{xy}\| = \|\overrightarrow{ts}\|.$$

   Express the statement that $ABCD$ is a square, using only the predicate P and logical conjunctions.

12. Quantifiers

There are two types of quantifiers.

1. **Universal quantifier** $\forall$: "for any", "for all", "for each", "every", ...
2. **Existential quantifier** $\exists$: "there exists", "for some", "for certain", "at least one", ...

In predicate logic, we can combine quantifiers with predicates to generate meaningful propositions. Suppose we have a domain of discourse D and a (unary) predicate P.

- When we apply the predicate to a constant term, we get a formula which bears a truth value.
- But if we apply the predicate to a variable, then we get a well-formed formula that does not bear any truth value, because of the free variable.
- That is where we use quantifiers: to bind the free variable and form a mathematical proposition.

Universal and Existential Statements

Universal statement: $\forall x\, P(x)$, "for all x, $P(x)$". The proposition is true if, whenever we replace x by any constant in the domain of discourse D, the resulting formula is always true, and it is false otherwise.

Existential statement: $\exists x\, P(x)$, "for some x, $P(x)$". The proposition is true if, when we replace x by some particular constant in D, the resulting formula is true, and it is false otherwise.

Negation of quantified statements.

1. $\neg(\forall x\, P(x)) \Leftrightarrow \exists x\, (\neg P(x))$.

   $\forall x\, P(x)$ is false when there exists one constant for which $P(x)$ is false, which is the same as saying that for some x, $P(x)$ is false.

2. $\neg(\exists x\, P(x)) \Leftrightarrow \forall x\, (\neg P(x))$.

   $\exists x\, P(x)$ is false when, no matter which constant we use, $P(x)$ is always false, which is the same as saying that for all x, "not $P(x)$" is a true statement.

Exercises

1. Let $S(x)$, $M(x)$ and $C(x)$ be predicates, where the domain is all people:

- $S(x)$: x is a student;
- $M(x)$: x studies mathematics;
- $C(x)$: x likes coding.

Translate the following English sentences into quantified statements.

(a) All students study maths.

(b) There is a student who studies maths and likes coding.

(c) Not every student studies maths.

(d) Not every student who studies maths likes coding, but every student who likes coding studies maths.

2. Let P, Q and R be predicates. Negate the following:

(a) $\forall x\, P(x)$

(b) $\forall x\, (\neg P(x) \to Q(x))$

(c) $\exists x\, (P(x) \vee (Q(x) \wedge R(x)))$

(d) $\forall x\, \exists y\, ((P(x) \wedge Q(y)) \to (Q(x) \vee \neg R(y)))$

(e) $\exists x\, (P(x) \to (\forall x\, (P(x) \to Q(x))))$

3. Determine if the following translation of natural language into a logical statement is correct.

"Everyone who is born in February naturally likes winter and hates summer."

Translation. Denote the following formulas:

$$B(x) : x \text{ is born in February}, \quad W(x) : x \text{ likes winter}, \quad S(x) : x \text{ hates summer}.$$

The original statement reads:

$$(\forall x\, (B(x) \to W(x))) \wedge (\forall x\, S(x)).$$

4. Let the universe D be the set of real numbers, and let A be an arbitrary subset of $\mathbb{R}$, with $A \neq \emptyset$. Show that

$$\forall x \in A,\ P(x) \Rightarrow \exists x \in A,\ P(x).$$

What if we drop the condition that $A \neq \emptyset$?

13. Rules of Inference

In predicate logic we have two types of statements.

1. **Instantiated statements**, such as $P(c)$.
   - They are achieved by applying a predicate to a constant in the domain of discourse.
   - Instantiated statements concern only a specified instance in the domain of discourse.
2. **Quantified statements**, such as $\forall x\, P(x)$ and $\exists x\, P(x)$.
   - They involve quantifiers and variables.
   - Each quantified statement is about what will happen as the variable ranges through the domain of discourse.

When the domain of discourse contains infinitely many elements, it is impossible for us to reconstruct a quantified statement using merely instantiated statements and connectives.

When we work on deductions, very often we have to carry out some operations, like adding two numbers $(2 + 3)$ or rotating a triangle. Most of these operations are applied directly to concrete instances instead of to the whole domain of discourse.

To facilitate these kinds of operations in predicate logic, we have some rules of inference that permit us to move back and forth between these two types of statements:

$$\underset{P(c)}{\text{instantiated statement}} \quad \begin{array}{c} \xleftarrow{\text{instantiation}} \\ \xrightarrow[\text{generalization}]{} \end{array} \quad \underset{\forall x\, P(x),\ \exists x\, P(x)}{\text{quantified statement}}$$

- **Instantiation** is deriving an instantiated statement from a quantified statement.
- **Generalization** is concluding a generalized statement from an instantiated statement.

1. **Universal instantiation.**

$$\frac{\forall x\, P(x)}{P(c)}, \qquad \text{for an arbitrary constant } c \text{ in the domain of discourse.}$$

We have the flexibility to choose any c of our preference.

2. **Existential instantiation.**

$$\frac{\exists x\, P(x)}{P(c)}, \qquad \text{for some fresh constant } c.$$

Here we have to use a c from the domain of discourse that has never been shown or used before.

3. **Universal generalization.**

$$\frac{P(c)}{\forall x\, P(x)}, \qquad c \text{ must be arbitrary.}$$

c should be a generic element: it should not have any property other than the fact that it is an element in D.

4. **Existential generalization.**

$$\frac{P(c)}{\exists x\, P(x)}, \qquad c \text{ can be specific.}$$

Here we can assume extra properties of c, as long as we know it is in D.

Example.

Every cat is lovely.	(1)
Chowi is a cat.	(2)
Therefore, Chowi is lovely.	(3)

Let $P(x)$: x is a cat, and $Q(x)$: x is lovely. Then

$$\begin{array}{ll} \forall x\,(P(x) \to Q(x)) & \\ \hline P(c) \to Q(c) & \text{universal instantiation} \\ \hline Q(c) & \text{modus ponens, with } P(c) \end{array}$$

where c is Chowi.

Example. *Socrates is a philosopher. Therefore, there exists a philosopher.*

This inference is not valid in propositional logic. In propositional logic, "Socrates is a philosopher" is treated as an atomic proposition, so we cannot access its internal structure, or reason about the existence of an individual with the property of being a philosopher.

The conclusion "there exists a philosopher" requires existential quantification, which is part of predicate logic rather than propositional logic. Therefore the inference cannot be justified within propositional logic. In predicate logic it follows by existential generalization.

Exercises

1. Determine if each of the following statements is true or false. Justify your answer.

(a) $\forall x \in \mathbb{R},\ x^2 = 1.$

(b) $\exists x \in \mathbb{R},\ x^2 = 1.$

2. Let the domain of discourse be all students in the university. Define

$$Q(x) : x \text{ is a Computer Science major,}$$
$$P(x) : x \text{ knows how to program in C.}$$

It is known that every student in the computer science major knows how to program in C, and there is someone who knows how to program in C. Dokja wants to prove $\exists x\, Q(x)$, and submitted the following proof:

> "Since there exists a student who knows C programming, let that student be a. Thus $P(a)$. Hence we know $Q(a) \Rightarrow P(a)$. Since $P(a)$ is true, it follows that $Q(a)$ must also be true. Therefore $\exists x\, Q(x)$."

Is the proof correct? Justify your answer.

14. Order of Quantifiers

In a compound statement that involves more than one quantifier:

1. if the quantifiers are the **same**, the order does not matter. For example,

$$\forall x\,\forall y\,P(x,y) \Leftrightarrow \forall y\,\forall x\,P(x,y) \qquad \text{and} \qquad \exists x\,\exists y\,P(x,y) \Leftrightarrow \exists y\,\exists x\,P(x,y);$$

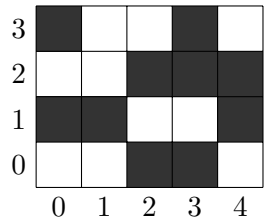
2. if the quantifiers are **different**, the order does matter. For example,

$$\forall x\,\exists y\,P(x,y) \not\Leftrightarrow \exists y\,\forall x\,P(x,y).$$

In $\forall x\,\exists y\,P(x,y)$, the choice of y can depend on x. In $\exists y\,\forall x\,P(x,y)$, the choice of y should not depend on x.

Example. Let $A = \{0,1,2,3,4\}$ and $B = \{0,1,2,3\}$. Consider the binary predicate

$$P(x,y) : \text{the square at } (x,y) \text{ is coloured black.}$$



- $\forall x \in A,\ \exists y \in B,\ P(x,y)$ means that each column has a black square. This is true in the picture.
- $\exists y \in B,\ \forall x \in A,\ P(x,y)$ means that there is a row that is entirely black. This is false in the picture.

Example. Determine if each of the following statements is true or false. Justify your answer.

(a) $\forall x \in \mathbb{Z},\ \exists y \in \mathbb{Z},\ 2 \mid x+y.$

(b) $\exists y \in \mathbb{Z},\ \forall x \in \mathbb{Z},\ 2 \mid x+y.$

Solution.

(a) *Claim:* the statement is true.

Proof. Let $x \in \mathbb{Z}$ be arbitrary. Pick $y = x$. Then $x + y = 2x$, that is, $2 \mid x+y$. □

(b) *Claim:* the statement is false.

Proof. We proceed by showing that its negation is true, that is,

$$\forall y \in \mathbb{Z},\ \exists x \in \mathbb{Z},\ 2 \nmid x+y.$$

To this end, let $y \in \mathbb{Z}$ be arbitrary. Pick $x = y+1$. Then $x + y = 2y+1$, that is, $2 \nmid x+y$. □

Exercises

1. Let P be a binary predicate with the formula $P(x, y)$ being $x = y$. Consider the following statements:

$$A : \forall x \, \forall y \, P(x, y),$$
$$B : \exists x \, \forall y \, P(x, y),$$
$$C : \exists x \, \exists y \, P(x, y).$$

Let the domain of discourse be $\mathbb{R}$.

(a) Is $A \to B$ true? How about its converse? Explain your answer.
(b) Is $B \to C$ true? How about its converse? Explain your answer.

2. Consider the following two formulas:

(a) $\forall x \, \exists y \, (x + y^2 > 2)$.
(b) $\exists x \, \forall y \, (x + y^2 > 2)$.

Let the domain of discourse be $\mathbb{R}$. Determine if each of them is true or false. Justify your answer.

3. Determine if the following statements are true or false.

(a) $\forall x \in \mathbb{R}, \ \exists y \in \mathbb{R} \ (x^2 = y)$.
(b) $\forall y \in \mathbb{R}, \ \exists x \in \mathbb{R} \ (x^2 = y)$.
(c) $\exists x \in \mathbb{R}, \ \forall y \in \mathbb{R} \ (x^2 = y)$.

Chapter 05 | Sets

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01. Set Equality

Definition: Even Integer

An integer n is called even if there exists an integer k such that $n = 2k$.

Exercises

1. Prove that $A = B$, where

$$A = \{m \in \mathbb{Z} \mid m = 2a \text{ for some integer } a\}$$

and

$$B = \{n \in \mathbb{Z} \mid n = 2b - 2 \text{ for some integer } b\}.$$

02. Product Sets

Product Set

Let A, B be two sets. The **product set** is defined as

$$A \times B = \{(x, y) \mid x \in A,\ y \in B\},$$

the set consisting of all ordered pairs of the form (x, y), where x belongs to A and y belongs to B.

Example. Let $A = \{x \mid x \text{ is a country}\}$ and $B = \{y \mid y \text{ is a city}\}$. Then

$$A \times B = \{(\text{USA}, \text{NYC}), (\text{India}, \text{Delhi}), (\text{Japan}, \text{London}), \ldots\}.$$

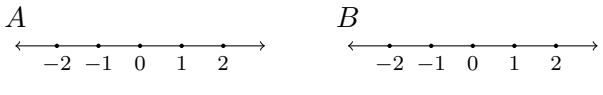
Remark. Let $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ be elements of $A \times B$. Then

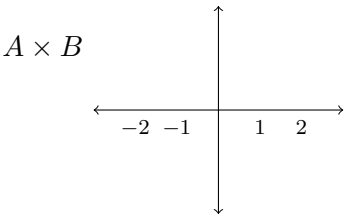
$$z_1 = z_2 \iff x_1 = x_2 \text{ and } y_1 = y_2.$$

Here we are talking about elements which are ordered pairs.

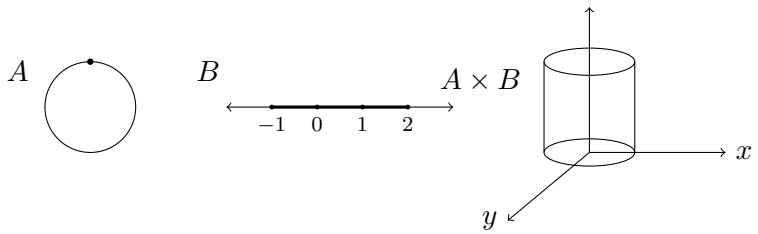
A product set can be visualised when A and B are both subsets of Euclidean spaces of low dimension.

Example. If A and B are both the real line, then $A \times B$ is the plane.





Example. If A is a circle and B is an interval of the real line, then $A \times B$ is a cylinder.



Exercises

1. Let $A = [0, 1] \cup \{2\}$ and $B = [2, 3] \subseteq \mathbb{R}$. Draw the set $A \times B$ on the Cartesian plane.

2. Determine if each of the following statements is true or false. Justify your answer.

(a) Let A, B be two sets. For any $C \subseteq A$, $D \subseteq B$, let $S = C \times D$. Then $C \times D \subseteq A \times B$.

(b) Let A, B be two sets. For any $S \subseteq A \times B$, there exist some $C \subseteq A$, $D \subseteq B$ such that $S = C \times D$.

3. For arbitrary sets A, B, C, determine if each of the following statements is true or false. Justify your answer.

(a) $(A \setminus B) \times C = (A \times C) \setminus (B \times C)$.

(b) $(A \setminus C) \times (B \setminus C) = A \times B \setminus (C \times C)$.

(c) $(A \times B) \times C = A \times (B \times C)$.

(d) If A and B are non-empty sets such that $A \times B = B \times A$, then $A = B$.

Chapter 06 | Relations

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01. Binary Relations

Binary Relation

Let A, B be two sets. A set $R \subseteq A \times B$ is called a **binary relation** from A to B. If $(x, y) \in R$, we say x is R**-related** to y, and we denote it by xRy.

A binary relation R from A to B is nothing more than a subset of the product set $A \times B$.

This definition not only agrees with our intuition of relations in our daily life, but also provides us with a mathematical language that is flexible and rigorous enough to describe relations among objects in two sets.

- $\emptyset$ is always called the **trivial relation**. It means "nobody in A is related to anybody in B".
- $A \times B$ is called the **universal relation**. It means "everyone in A is related to everyone in B".

Example. Define $R \subseteq \mathbb{Z} \times \mathbb{Z}$ by

$$\forall x, y \in \mathbb{Z}, \quad xRy \ := \ 2 \mid x - y.$$

So x is related to y if and only if x and y have the same parity.

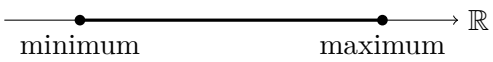
This relation performs a *classification*: based on it, we can put each integer into the set of even integers or the set of odd integers.

Example. Define $R \subseteq \mathbb{R} \times \mathbb{R}$ by

$$\forall x, y \in \mathbb{R}, \quad xRy \ := \ x \leq y.$$

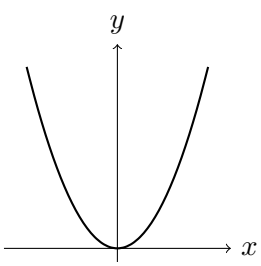
So x is related to y if and only if x is less than or equal to y.

This relation establishes an *order*. With it we can also discuss the minimum and maximum of a given subset of $\mathbb{R}$.



Example. Define $R \subseteq \mathbb{R} \times \mathbb{R}$ by

$$\forall x, y \in \mathbb{R}, \quad xRy \ := \ y = x^2.$$



This graph is a subset of $\mathbb{R}^2$, and hence it can naturally be considered as a binary relation from $\mathbb{R}$ to itself. In this way binary relations *define functions*: defining a function is a special case of defining a binary relation.

Exercises

1. Read the following definitions and answer the question below.

- **Inverse of a relation:** let R be a relation from A to B. The *inverse* of R is the relation from B to A defined by
$$R^{-1} = \{(b, a) \in B \times A : (a, b) \in R\}.$$
- **Composition of relations:** let R be a relation from A to B, and let S be a relation from B to C. The *composition* of R and S is the relation from A to C defined by
$$S \circ R = \{(a, c) \in A \times C : \exists b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}.$$

Let $A = \{1, 2, 3\}$, $B = \{a, b, c\}$, $C = \{x, y\}$, and define

$$R = \{(1, a), (1, b), (2, c)\} \subseteq A \times B, \qquad S = \{(a, x), (b, x), (c, y)\} \subseteq B \times C.$$

Compute R^{-1} and $S \circ R$.

2. Let R be a relation from A to B, S a relation from B to C, and T a relation from C to D. Prove that composition is associative:
$$T \circ (S \circ R) = (T \circ S) \circ R.$$

3. Let R and S be relations from A to B. Prove that
$$(R \cap S)^{-1} = R^{-1} \cap S^{-1}.$$

02. Equivalence Relations

Equivalence Relation

Let R be a binary relation on a set A. We say that R is

- **reflexive** if $\forall x \in A$, xRx;
- **symmetric** if $\forall x, y \in A$, $xRy \Rightarrow yRx$;
- **transitive** if $\forall x, y, z \in A$, $(xRy \wedge yRz) \Rightarrow xRz$.

If R satisfies all three properties above, then R is called an **equivalence relation** on A.

An equivalence relation R is often denoted by $\sim$. An ordered pair $(x, y) \in R$ is written as $x \sim y$ (x is related to y).

Example (1). Let $\sim$ be a binary relation on $\mathbb{R}$ defined by

$$\forall x, y \in \mathbb{R}, \quad x \sim y \;:=\; x = y.$$

Here our set R is called the **diagonal set**, which consists of the ordered pairs whose two coordinates are identical:

$$R = \{(x, x) \mid x \in \mathbb{R}\} \subseteq \mathbb{R}^2.$$

Example (2). Let $\sim$ be a binary relation on $\mathbb{R}$ defined by

$$\forall x, y \in \mathbb{R}, \quad x \sim y \;:=\; |x| = |y|.$$

- Reflexivity: $\forall x \in \mathbb{R}$, $|x| = |x|$.
- Symmetry: $\forall x, y \in \mathbb{R}$, $|x| = |y| \Rightarrow |y| = |x|$.
- Transitivity: $\forall x, y, z \in \mathbb{R}$, $(|x| = |y|) \wedge (|y| = |z|) \Rightarrow |x| = |z|$.

So $\sim$ is an equivalence relation.

Example (3). Let $\sim$ be a binary relation on $\mathbb{R}$ defined by

$$\forall x, y \in \mathbb{R}, \quad x \sim y \ := \ (xy > 0) \vee (x = y = 0).$$

- Reflexivity: let $x \in \mathbb{R}$.
  - Either $x = 0$, so $x = x = 0$;
  - or $x \neq 0$, and hence $x^2 > 0$.
- Symmetry: let $x, y \in \mathbb{R}$. If $x \sim y$, then
  - either $x = y = 0$, that is, $y = x = 0$;
  - or $xy > 0$, and hence $yx > 0$.
- Transitivity: let $x, y, z \in \mathbb{R}$. If $(x \sim y) \wedge (y \sim z)$, then
  - either $y = 0$, in which case $x = z = 0$;
  - or $y \neq 0$, in which case $xy > 0$ and $yz > 0$, which in turn implies $xy^2z > 0$, and therefore $xz > 0$.

So $\sim$ is an equivalence relation.

Example (4). Let $\sim$ be a binary relation on $\mathbb{R}$ defined by

$$\forall x, y \in \mathbb{R}, \quad x \sim y.$$

That is, every real number is related to every real number. This is an equivalence relation as well.

Exercises

1. Goofus and Gallant are learning equivalence relations. Goofus claims that the condition of reflexivity is redundant: he believes every relation which is symmetric and transitive is reflexive. Gallant disagrees. Goofus gives the following proof:

> **Proof.** Let $\sim$ on a set A be symmetric and transitive. Let $a \in A$ be arbitrary and let $b \in A$ be an arbitrary element distinct from a, with $a \sim b$. By symmetry, $b \sim a$. By transitivity,
>
> $$(a \sim b) \wedge (b \sim a) \Rightarrow a \sim a.$$
>
> Thus reflexivity is proved.

Cite the line(s) where Goofus' proof fails and explain why Goofus' proof is incorrect.

2. Suppose that $\mathcal{R}$ is a relation on a set S. Define the relation

$$\mathcal{R}^* = \{(x, y) \in S \times S : (y, x) \in \mathcal{R}\}.$$

Determine whether $\mathcal{R} \cup \mathcal{R}^*$ is necessarily an equivalence relation. Justify your answer.

3. For each of the following relations, determine if it is an equivalence relation or not, and explain your claim in detail. In case it is an equivalence relation, describe the quotient space.

(a) The relation $\sim$ on the set of integers $\mathbb{Z}$ defined by

$$\forall a, b \in \mathbb{Z}, \quad a \sim b := 4 \mid (a + b).$$

(b) The relation $\sim$ on the set Σ^* of finite strings over the alphabet $\{a, b\}$ defined by

$$\forall s, t \in \Sigma^*, \quad s \sim t := s \text{ and } t \text{ have the same number of } a\text{'s and the same number of } b\text{'s.}$$

(c) The relation $\sim$ on the set of real numbers $\mathbb{R}$ defined by

$$\forall x, y \in \mathbb{R}, \quad x \sim y := \lfloor x \rfloor = \lfloor y \rfloor.$$

(d) The relation $\sim$ on the set of integers $A = \{2^n \mid n \in \mathbb{N}\}$ defined by

$$\forall a, b \in A, \quad a \sim b := a \mid b \vee b \mid a.$$

(e) The relation $\sim$ on the set of real numbers $\mathbb{R}_*$ defined by

$$\forall a, b \in \mathbb{R}_*, \quad a \sim b := \frac{a}{b} \in \mathbb{Q}.$$

(f) The relation $\sim$ on the set of polynomials $\mathbb{R}[x]$ with real coefficients defined by

$$\forall p, q \in \mathbb{R}[x], \quad p \sim q := p - q \text{ is divisible by a fixed polynomial } d(x).$$

03. Equivalence Classes

Equivalence Class

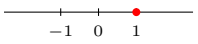
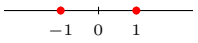
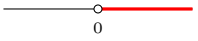
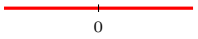
Let $\sim$ be an equivalence relation on a set A. For any element $x \in A$, the set

$$[x] := \{y \in A \mid x \sim y\}$$

is called the **equivalence class** represented by x.

$[x]$, in general, is not an element of A, but a subset of A.

Example. Find the equivalence class [1] for each of the following equivalence relations on $\mathbb{R}$.

Relation	Equivalence class [1]	Visualization
$x = y$	$\{1\}$	
$\lvert x \rvert = \lvert y \rvert$	$\{-1, 1\}$	
$(xy > 0) \vee (x = y = 0)$	$\mathbb{R}_{>0}$	
all $x, y \in \mathbb{R}$ related	$\mathbb{R}$	

Assume that we have a set A equipped with an equivalence relation, and consider the equivalence class represented by x, as well as that represented by y. What can we say about these as subsets of A? The answer lies in the following two propositions.

Proposition 1

Let A be a non-empty set and $\sim$ be an equivalence relation on A. Then

$$\forall x, y \in A, \quad x \sim y \iff [x] = [y].$$

Proof. ($\Rightarrow$) Assume that $x \sim y$. Let $z \in [y]$ be arbitrary, that is, $y \sim z$. By transitivity, it follows that $x \sim z$, so $z \in [x]$. Hence $[y] \subseteq [x]$. Since $y \sim x$ by symmetry, a similar argument shows that $[x] \subseteq [y]$. We conclude that $[x] = [y]$.

($\Leftarrow$) Assume that $[x] = [y]$. By reflexivity, we have $y \in [y]$. Thus $y \in [x]$, which means that $x \sim y$. $\square$

Proposition 2

Let A be a non-empty set and $\sim$ be an equivalence relation on A. Then

$$\forall x, y \in A, \quad x \not\sim y \iff [x] \cap [y] = \emptyset.$$

Proof. ($\Rightarrow$) Assume that $x \not\sim y$. Suppose, to the contrary, that $z \in [x] \cap [y]$, that is, $x \sim z$ and $y \sim z$. By symmetry, $z \sim y$, and by transitivity it follows that $x \sim y$, which is a contradiction. We conclude that $[x] \cap [y] = \emptyset$.

($\Leftarrow$) Assume that $[x] \cap [y] = \emptyset$. Suppose, to the contrary, that $x \sim y$. Then $y \in [x]$. Moreover, $y \in [y]$ by reflexivity. It turns out that $y \in [x] \cap [y]$, which is a contradiction. Thus $x \not\sim y$. □

Exercises

1. Let R and S be equivalence relations on a set A. Given any element $a \in A$, write $[a]_R$, $[a]_S$ for the classes of a under R and S.

(a) Prove that $R \cap S$ is an equivalence relation on A.

(b) Prove that for every $a \in A$, $[a]_{R \cap S} = [a]_R \cap [a]_S$.

(c) Construct some equivalence relations R and S on $A = \{1, 2, 3\}$ such that $R \cup S$ is **not** an equivalence relation on A.

2. Let R and S be equivalence relations on a set A.

(a) Prove that $R \subseteq S$ if and only if

$$\forall a \in A, \quad [a]_R \subseteq [a]_S.$$

(b) Prove that if $R \subseteq S$, then

$$\forall a \in A, \quad [a]_S = \bigcup_{x \in [a]_S} [x]_R.$$

3. Let R and S be equivalence relations on a set A. Prove that the composition of relations $R \circ S$ is an equivalence relation if and only if $R \circ S = S \circ R$.

04. Quotient Sets

Quotient Set

Let A be a non-empty set equipped with an equivalence relation $\sim$. The **quotient set** of A by $\sim$, denoted by

$$A/\!\sim := \{[x] \mid x \in A\},$$

is the set of all equivalence classes of A under $\sim$.

Each element of $A/\!\sim$ is an equivalence class of A, that is, a subset of A. In other words, $A/\!\sim \subseteq \mathcal{P}(A)$.

Example. Let $\sim$ be the equivalence relation on $\mathbb{Z}$ where

$$\forall x, y \in \mathbb{Z}, \quad x \sim y \ := \ 2 \mid x - y.$$

Then

$$[0] = \{\ldots, -6, -4, -2, 0, 2, 4, 6, \ldots\},$$
$$[1] = \{\ldots, -5, -3, -1, 1, 3, 5, \ldots\}.$$

The quotient set consists of two elements:

$$\mathbb{Z}/\!\sim := \{[x] \mid x \in \mathbb{Z}\} = \{[0], [1]\}.$$

Here [0] and [1]

1. are both non-empty,
2. are disjoint, and
3. satisfy $[0] \cup [1] = \mathbb{Z}$.

Exercises

1. Let $A = \mathcal{P}(\{1,2,3\})$ be the power set of $\{1,2,3\}$, and define a relation $\sim$ on A, where

$$\forall X, Y \in A, \quad X \sim Y := |X| = |Y|;$$

here $|X|$ stands for the number of elements in X.

(a) Prove that $\sim$ is an equivalence relation on A.

(b) Write down the equivalence class $[\{1\}]$.

(c) Write down the quotient set $A/\sim$ explicitly. How many elements does it have?

2. Let R and S be equivalence relations on a set A with $R \subseteq S$. Define a relation $\bar{S}$ on the quotient set A/R by

$$([a]_R, [b]_R) \in \bar{S} \iff (a,b) \in S.$$

(a) Prove that this definition does not depend on the choice of representatives: if $[a]_R = [a']_R$ and $[b]_R = [b']_R$, then $(a,b) \in S$ if and only if $(a',b') \in S$.

(b) Prove that $\bar{S}$ is an equivalence relation on A/R.

(c) Prove that $[[a]_R]_{\bar{S}}$ is $\{[x]_R : x \in [a]_S\}$, and that the union of the elements in this class is exactly $[a]_S$.

3. Let R and S be equivalence relations on a set A.

(a) Prove that $A/R = A/S$ if and only if $R = S$.

(b) Prove that

$$A/(R \cap S) = \{X \cap Y : X \in A/R,\ Y \in A/S,\ X \cap Y \neq \emptyset\}.$$

05. Partitions

Partition

Let A be a non-empty set. A set of its subsets

$$\mathcal{P} = \{U_i \in \mathcal{P}(A) \mid i \in I\},$$

where I is an index set, is called a **partition** of A if

1. $\forall i \in I,\ U_i \neq \emptyset$ (non-emptiness)
2. $\forall i, j \in I,\ i \neq j \Rightarrow U_i \cap U_j = \emptyset$ (pairwise disjointness)
3. $\bigcup_{i \in I} U_i = A$ (cover)

This definition simply means that we cut the set A into different components, and each component U_i in this partition is called a **cell** of the partition.

Theorem: Partition by Equivalence Classes

Let A be a non-empty set and $\sim$ be an equivalence relation on A. Then its quotient set $A/\sim$ forms a partition of A.

Proof. Recall that $A/\sim = \{[x] \mid x \in A\}$.

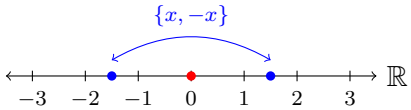
1. *Non-emptiness.* Let $[x]$ be an arbitrary equivalence class, represented by x. Then by reflexivity, $x \in [x]$.
2. *Pairwise disjointness.* Let $[x], [y]$ be two arbitrary distinct equivalence classes, represented by x and y respectively. Since $[x] \neq [y]$, Proposition 1 gives $x \not\sim y$, and then Proposition 2 gives $[x] \cap [y] = \emptyset$.
3. *Cover.* We prove $\bigcup_{x\in A}[x] = A$ by double set inclusion.
   - $A \subseteq \bigcup_{x\in A}[x]$: let $x \in A$ be arbitrary. By reflexivity, $x \in [x]$, so x is in the union.
   - $\bigcup_{x\in A}[x] \subseteq A$: since $[x] \subseteq A$ for every $x \in A$, so is their union.

□

Example. Consider the partition by equivalence classes for each of the following equivalence relations on $\mathbb{R}$.

1. $\forall x, y \in \mathbb{R}$, $x \sim y := x = y$. Then
$$\mathbb{R}/\sim = \{\{x\} \mid x \in \mathbb{R}\}.$$
So the quotient set $\mathbb{R}/\sim$ is the set of all equivalence classes $[x]$, each of which has only one element, $x \in [x]$.
2. $\forall x, y \in \mathbb{R}$, $x \sim y := |x| = |y|$. Then
$$\mathbb{R}/\sim = \{\{x, -x\} \mid x \in [0, \infty)\}.$$
Here the quotient set consists of the singleton set $\{0\}$ together with all the pairs $\{x, -x\}$ over the positive real numbers.
3. $\forall x, y \in \mathbb{R}$, $x \sim y := (xy > 0) \vee (x = y = 0)$. Then
$$\mathbb{R}/\sim = \{\{0\}, (0, \infty), (-\infty, 0)\}.$$
Here we have three elements under this equivalence relation: $\{0\}$, $(0, \infty)$ and $(-\infty, 0)$.
4. $\forall x, y \in \mathbb{R}$, $x \sim y$. Then
$$\mathbb{R}/\sim = \{\mathbb{R}\}.$$
Under this equivalence relation there is only one equivalence class, that is, the set $\mathbb{R}$ itself.

The cells of the partition in item 2, on the real line:



It is interesting to note that if we were given a partition of the set A in the first place, we could then construct an equivalence relation. If $\mathcal{P} = \{U_i \in \mathcal{P}(A) \mid i \in I\}$ is a partition of A, then the binary relation

$$\forall x, y \in A, \quad x \sim y \;:=\; \exists i \in I,\; x, y \in U_i$$

is an equivalence relation on A.

06. Partial Orders

Partial orders are another kind of binary relation.

Partial Order

Let R be a binary relation on a set A. We say that R is

- **reflexive** if $\forall x \in A$, xRx;
- **anti-symmetric** if $\forall x, y \in A$, $(xRy) \wedge (yRx) \Rightarrow x = y$;
- **transitive** if $\forall x, y, z \in A$, $(xRy) \wedge (yRz) \Rightarrow xRz$.

If R satisfies all three properties above, then R is called a **partial order** on A.

A partial order R is often denoted by $\preceq$. An ordered pair $(x, y) \in R$ is written as $x \preceq y$: x precedes or is equal to y.

Example. Consider the binary relation $\preceq$ defined on $\mathbb{N}$ by

$$\forall x, y \in \mathbb{N}, \quad x \preceq y \ := \ x \leq y.$$

- Reflexivity: for all $x \in \mathbb{N}$, we have $x \leq x$, hence $x \preceq x$.
- Anti-symmetry: for all $x, y \in \mathbb{N}$, if $x \preceq y$ then $x \leq y$, and if $y \preceq x$ then $y \leq x$. Together these give $x = y$.
- Transitivity: for all $x, y, z \in \mathbb{N}$, if $x \preceq y$ then $x \leq y$, and if $y \preceq z$ then $y \leq z$. So $x \leq z$, and hence $x \preceq z$.

Example. Consider the binary relation $\preceq$ defined on $\mathbb{N}$ by

$$\forall x, y \in \mathbb{N}, \quad x \preceq y \ := \ x \mid y.$$

- Reflexivity: for all $x \in \mathbb{N}$, we have $x \mid x$, hence $x \preceq x$.
- Anti-symmetry: for all $x, y \in \mathbb{N}$,

  $$x \preceq y \Rightarrow x \mid y \Rightarrow \exists p \in \mathbb{N}, \ y = px,$$
  $$y \preceq x \Rightarrow y \mid x \Rightarrow \exists q \in \mathbb{N}, \ x = qy.$$

  Then $y = (pq)y$, so $(x = y = 0) \vee (p = q = 1)$, and in both cases $x = y$.
- Transitivity: for all $x, y, z \in \mathbb{N}$,

  $$x \preceq y \Rightarrow x \mid y \Rightarrow \exists p \in \mathbb{N}, \ y = px,$$
  $$y \preceq z \Rightarrow y \mid z \Rightarrow \exists q \in \mathbb{N}, \ z = qy.$$

  Then $z = (pq)x$, so $x \mid z$, that is, $x \preceq z$.

Exercises

1. For each of the following relations $\mathcal{R}$ on $A = \{1, 2, 3\}$, determine whether it is a partial order. Justify your answer.

(a) $\mathcal{R} = \{(1,1), (2,2), (3,3), (1,2), (2,3)\}$.
(b) $\mathcal{R} = \{(1,1), (2,2), (3,3), (1,2), (2,1)\}$.

2. For each of the following relations $\mathcal{R}$ on $\mathbb{Z}$, determine whether it is a partial order. Justify your answer.

(a) The binary relation $\mathcal{R}$ given by

$$\forall a, b \in \mathbb{Z}, \quad a \mathcal{R} b := a \mid b.$$

(b) The binary relation $\mathcal{R}$ given by

$$\forall a, b \in \mathbb{Z}, \quad a \mathcal{R} b := \forall p \in \mathbb{N},\ (p \text{ is prime}) \wedge (p \mid a) \Rightarrow p \mid b.$$

3. For each of the following relations $\mathcal{R}$ on $\mathcal{P}(\mathbb{N})$, determine whether it is a partial order. Justify your answer.

(a) The binary relation $\mathcal{R}$ given by

$$\forall A, B \in \mathcal{P}(\mathbb{N}), \quad A \mathcal{R} B := A \setminus B \text{ is a finite set.}$$

(b) The binary relation $\mathcal{R}$ given by

$$\forall A, B \in \mathcal{P}(\mathbb{N}), \quad A \mathcal{R} B := A \subseteq B \text{ and } A \setminus B \text{ is a finite set.}$$

07. Total Orders

Exercises

1. Let $S = \{1, 2, \ldots, 2025\}$, and consider the power set $\mathcal{P}(S)$ (the set of all subsets of S). Given two subsets $A, B \subseteq S$, we define the symmetric difference of A and B as

$$A \triangle B := (A \cup B) \setminus (A \cap B).$$

We denote by $x_{AB} = \max_{x \in A \triangle B} x$ the maximum element of $A \triangle B$.

(a) Prove that $A \triangle B = (A \setminus B) \cup (B \setminus A)$.
(b) Let $C = \{1, 2, 3, 5\}$ and $D = \{2, 4, 5, 9\}$. Compute $C \triangle D$ and x_{CD}.
(c) Consider the relation $\preceq_1$ on $\mathcal{P}(S)$ defined by

$$\forall A, B \in \mathcal{P}(S), \quad A \preceq_1 B := A \subseteq B.$$

Is $\preceq_1$ a partial order on $\mathcal{P}(S)$? Is it a total order? Explain your answer in detail.
(d) Consider the relation $\preceq_2$ on $\mathcal{P}(S)$ defined by

$$\forall A, B \in \mathcal{P}(S), \quad A \preceq_2 B := x_{AB} \in B \text{ or } A \triangle B = \emptyset.$$

Is $\preceq_2$ a partial order on $\mathcal{P}(S)$? Is it a total order? Explain your answer in detail.

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01. Functions

Exercises

1. Let A be a set consisting of m elements and B a set consisting of n elements. How many different functions are there from A to B?

2. For each of the following two binary relations from $\mathbb{N}$ to $\mathbb{N}$, determine if it is a function or not. Justify your answer.

(a) $\mathcal{R}_1 = \{(1,2),(2,4),(3,6),(4,8)\}$.

(b) $\mathcal{R}_2 = \{(1,5),(2,7),(1,9),(3,11)\}$.

3. Is the binary relation

$$\mathcal{R} = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$

a function $f : [-1,1] \to \mathbb{R}$? Justify your answer.

4. Let $A \neq \emptyset$ and $B, C \subseteq A$. Consider the indicator functions

$$1_B : A \to \{0,1\}, \qquad 1_C : A \to \{0,1\}.$$

Is $1_B \cdot 1_C$ an indicator function? If yes, then it is the indicator function of which set?

5. Do there exist two functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ such that the following two conditions

- neither f nor g is a constant function, and
- the composition $f \circ g$ is a constant function

are satisfied at the same time? Justify your answer.

02. Image and Preimage

Exercises

1. Construct a function $f : \mathbb{N} \to \mathbb{N}$ such that $f^{-1}(\{n\})$ contains precisely n elements for each $n \in \mathbb{N}$.

2. Consider the function $f : (0, \infty) \to \mathbb{R}$ given by

$$f(x) = x - \frac{1}{x}.$$

Find the range of f. Justify your answer.

3. Let $A \subseteq \mathbb{R}$ be a symmetric set, in the sense that

$$\forall x \in \mathbb{R},\ x \in A \to -x \in A.$$

Let $f : A \to \mathbb{R}$ be an odd function. Determine if each of the following statements is true or false. Justify your answer.

(a) $0 \in f(A)$.
(b) If $0 \in A$, then $0 \in f(A)$.
(c) If $0 \notin A$, then $0 \notin f(A)$.
(d) If $0 \in f(A)$, then $f^{-1}(\{0\}) = \{0\}$.

03. Properties of Images and Preimages

Exercises

1. Let $f : A \to B$. Prove that

$$f(A_1 \cap f^{-1}(B_1)) = f(A_1) \cap B_1$$

for every $A_1 \subseteq A$ and $B_1 \subseteq B$.

2. Recall the definition of a partition.

Definition. Let A be a set. A *partition* of A is a collection $\mathcal{P}$ of subsets of A such that:

(a) $\forall P \in \mathcal{P}$, $P \neq \emptyset$;
(b) $\bigcup_{P \in \mathcal{P}} P = A$;
(c) $\forall P, Q \in \mathcal{P}$, either $P = Q$ or $P \cap Q = \emptyset$.

Let $f : A \to B$. Determine if each of the following statements is true or false. Justify your answer.

(a) If $\{C_i\}_{i \in I}$ is a partition of A, then $\{f(C_i)\}_{i \in I}$ is a partition of B.
(b) If $\{D_i\}_{i \in I}$ is a partition of B, then $\{f^{-1}(D_i)\}_{i \in I}$ is a partition of A.

3. Let $f : A \to B$ be a function. Prove that

$$\forall V \subseteq B, \quad f(f^{-1}(f(f^{-1}(V)))) = f(f^{-1}(V)),$$
$$\forall U \subseteq A, \quad f^{-1}(f(f^{-1}(f(U)))) = f^{-1}(f(U)).$$

04. Injectivity, Surjectivity and Bijectivity

Exercises

1. Let $f, g : \mathbb{R} \to \mathbb{R}$ be two functions. Determine if each of the following statements is true or false. Justify your answer.

(a) If either f or g is not injective, then $f + g$ is not injective.
(b) If both f and g are injective, then $f + g$ is injective.
(c) If either f or g is not surjective, then $f + g$ is not surjective.
(d) If both f and g are surjective, then $f + g$ is surjective.

2. Let A, B, C be sets. Let $f : A \to B$ and $g : B \to C$ be two functions. By $h = g \circ f$ we denote their composition. Determine if each of the following statements is true or false. Justify your answer.

(a) If f and g are both injective, then h is injective.
(b) If f and g are both surjective, then h is surjective.
(c) If f and g are both bijective, then h is bijective.
(d) If f is injective, and g is surjective, then h is bijective.
(e) If f is surjective, and g is injective, then h is bijective.
(f) If h is injective, then both f and g are injective.
(g) If h is surjective, then both f and g are surjective.
(h) If h is bijective, then both f and g are bijective.
(i) If h is bijective, then f is injective and g is surjective.
(j) If h is bijective, then f is surjective and g is injective.

3. Let $f : A \to B$ be a function. Prove that f is bijective if and only if

$$\forall U \subseteq A, \quad f(A \setminus U) = B \setminus f(U).$$

4. Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f^{(n)}(x) = f(x)$ for some $n \geq 2$, where $f^{(n)}$ denotes the composition of f with itself n times. Determine if each of the following statements is true or false. Justify your answer.

(a) If f is injective, then f is bijective.
(b) If f is surjective, then f is bijective.

05. Cardinality

Exercises

1. Show that the set of even numbers E and the set of odd numbers O have the same cardinality.

2. Prove that if $f : A \to B$ is an injection, then there exists an injection $F : \mathcal{P}(A) \to \mathcal{P}(B)$. Based on this, conclude that

$$|A| = |B| \Rightarrow |\mathcal{P}(A)| = |\mathcal{P}(B)|.$$

3. Let $a, b \in \mathbb{R}$ with $a < b$. Show that

$$|[a, b]| = |(a, b)| = |[a, b)| = |(a, b]| = |\mathbb{R}|.$$

4. Prove that if A is countable, then $|\mathbb{R} \setminus A| = |\mathbb{R}|$.

06. The Cantor–Schröder–Bernstein Theorem

Exercises

1. Prove that $[0,1] \cup [2,3]$ has the same cardinality as $[-1,1]$.

2. Prove that if $|A| = |B|$, then $|\mathcal{P}(A)| = |\mathcal{P}(B)|$, using the Cantor–Schröder–Bernstein theorem.

07. Countable Sets

Exercises

1. Discuss whether each of the following operations produces a finite set or a countably infinite set.

(a) A finite union of finite sets.
(b) A countable union of finite sets.
(c) A finite union of countable sets.
(d) A countable union of countable sets.

Justify your answer.

2. In this exercise we consider the cardinality of two countable sets.

(a) Prove that the function

$$g : \mathbb{N} \times \mathbb{N} \to \mathbb{N}, \qquad g(m,n) = \frac{(m+n)(m+n+1)}{2} + n$$

is bijective.
(b) Prove that $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$.
(c) Prove that the Cartesian product of two countable sets is countable.

08. $\mathbb{Q}$ Is Countable

Exercises

1. In this exercise, we use another approach to prove that $\mathbb{Q}$ is countable. Recall that

$$\mathbb{Q} = \left\{ x = \frac{p}{q} \;\middle|\; p, q \in \mathbb{Z},\ q \neq 0 \right\}.$$

To this end, let $\mathbb{Q} = \mathbb{Q}_+ \cup \mathbb{Q}_- \cup \{0\}$, where $\mathbb{Q}_+$ and $\mathbb{Q}_-$ are the sets of positive rational numbers and negative rational numbers, respectively. We first show that $\mathbb{Q}_+$ is countable. Note that

$$\mathbb{Q}_+ = \left\{ x = \frac{p}{q} \;\middle|\; p, q \in \mathbb{N},\ p, q > 0 \right\}.$$

Assume without loss of generality that p, q are moreover coprime, that is, p, q have no common integer factors other than 1 (this means that $2/4$ should be denoted as $1/2$, and $10/15$ as $2/3$, to name but a few). Consider, for any $k \in \mathbb{N}$, $k \geq 2$,

$$A_k = \left\{ x = \frac{p}{q} \in \mathbb{Q}_+ \;\middle|\; p + q = k \right\}.$$

(a) List the elements in A_2, A_3, A_4.
(b) Prove that $\forall k \in \mathbb{N}$, $k \geq 2$, A_k is a finite set.
(c) Prove that $\forall i, j \in \mathbb{N}$, $i, j \geq 2$, $i \neq j$, $A_i \cap A_j = \emptyset$.
(d) Prove that $\mathbb{Q}_+ = \bigcup_{i=2}^{\infty} A_i$. Based on that, prove that $\mathbb{Q}_+$ is countable.
(e) Prove that $|\mathbb{Q}_+| = |\mathbb{Q}_-|$.
(f) Prove that $\mathbb{Q}$ is countable.